

smart, so it's not as simple as an anti-virus saying, "Aha, a virus; [Delete], problem solved!" Think of it more like an argument between two sentient beings-it could rage forever! Now, consider the one lone sentry (anti-virus software) on your computer, battling all the viruses that the world throws at you. Trust me, your anti-virus will lose, or at least use up all your resources while fighting them off!

The Borg!

Like viruses do today, what if a smart-virus attack starts taking over computers? How do we stop them? Since we're assuming AI is possible, formatting a PC and ~killing~ the AI software on our computers might not be as easy as it is today! Will infected computers band together to form a real-life version of the ~Borg~ from ~Star Trek~?

What intrigues me more is how the "dumb" viruses of today behave:

Dumb, Not Dumber

Billions of US dollars are lost on attacks by viruses on important data, not to mention the millions of man hours lost rectifying the problems. Consider how these viruses are designed: they're built to attack, hide, replicate and spread like wildfire! They faithfully mimic their biological counterparts, and are just as destructive.

So, viruses are the first software programs to exhibit signs of life. They reproduce, attack, defend and hide - much like we would when at war (well, maybe not reproduce, but you get the general idea!). The other side of the argument states that on their own, viruses are useless, and need a host PC/operating system/software/EXE, whatever..., in order to function. This debate is strikingly analogous to the ones being fought for decades now about the "alive or not" status of biological viruses.

What about AI assisted anti-virus software then? Would they be alive? Perhaps, but anti-virus software have the distinction of being defenders. Think about how your computer behaves when it's infected: first the virus attacks the anti-virus software, disabling it; then it takes over the system, stopping all software and services that could be a threat to it. Now, once it's in control of the system,

it prevents your attempts to restart your anti-virus or virus cleaning software. What that one virus has done, is attacked you covertly, vanquished your guards, killed all the men in the castle (PC) and then reproduced, using your resources to make more of itself. Then it's taken guard in your own watchtowers, using your own weapons against any attempt by you to reclaim your castle. The virus has reversed roles from attack to defence! No other software is programmed this way, simply because they do not need to be. It doesn't change the fact that they're behaving very similar to the way we do!

Bicentennial Man

Like in the movie or book - take your pick - where an artificially intelligent robot with a positronic brain fights for the right to be called human. So what happens if AI software wants to be considered a living entity? What if they start demanding a regular eight-hour workday? What if they form unions and decide to go on strike demanding better work conditions-like higher-end hardware? This may sound far-fetched, but remember, we're talking intelligent software here, with a sense of self!

If the computing world is equated to ours, viruses are the extremists and terrorists - very unpleasant characters, but a part of "society" nonetheless! So although AI could bring about a lot of good, even with all software and hardware programmed with Asimov's famed "three laws of robotics" in mind, viruses will not be governed by such limitations! Will a new generation of terror be brought to our lives through our fattening broadband pipes?

Getting Back

But let's not digress from the topic at hand. I'm still left wondering if viruses (or even software, for that matter) can or will be considered living entities. I must accentuate the fact that I'm not necessarily talking about right now, but rather about the day when a group of researchers in an inconspicuous lab (some time in the not-so-distant future) cheer, pat each other on the backs, pop open a bottle of some exotic wine and celebrate the birth of a digitally sustainable intelligence that lives on electric-

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ity and can learn on its own!

I'm not worried about whether humans will accept such "programs" as living entities or not; I'm more concerned about whether we will have a choice but to do so!

The world's economy, our personal savings account, most of our records, and a lot more, are all dependant on digital data. If stock market servers crash (or refuse to co-operate), if your bank loses all your data, and if all computers just stopped working, we'd be facing an apocalypse of sorts! I'd rather not lose my life savings to a computer bug, thank you!

Paranoia

Yes, there seems to be a lot of that recently, and my article in last month's ~Digit~ was against PC-related paranoia. Although I still think it's pointless to be paranoid about technology in today's world, I'm unsure of what the future holds. As long as we don't depended on technology and software to think for us, we have absolutely nothing to worry about - let's hope countries with nuclear weapons don't leave computers in charge of the "big red buttons."

A ~Terminator~-Induced Phobia? Absolutely!

So Are Viruses Alive?

The fact is, religious beliefs about souls and spirits aside, computers and the software within them have the basic pre-requisites for life. Will this era be considered the primordial soup from which digital life was born?

I think the evolution of software into thinking entities is just a natural progression! What I'm afraid of is that viruses, which are already mimicking sentient, self-preserving beings, will get there first!

Will we soon see "e-telligence"?

More importantly, do we want to? ■

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